ICDAR 2026 Competition on Long-Term Handwriting Author Identification

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Abstract. Identifying authorship in historical documents is a fundamental task in the context of historical document analysis. Author retrieval and writer identification have received considerable attention from the ICDAR community, as both pattern recognition and transductive capabilities are essential for time-efficient retrieval schemes. However, current approaches have not addressed the temporal dimension of authorship: no existing benchmark tests whether intelligent reading systems can identify common authorship across manuscripts separated by decades. This limitation reduces the real-world effectiveness of writer identification technologies, particularly for writers with sparse manuscript production or limited surviving samples. To circumvent this limitation, we propose the AnyScript dataset, which is divided into two tasks: First, participants receive query pages from a book and must identify the remaining pages of that book in the competition dataset ("`pages`" task). Second, participants must retrieve pages from the same book and also books by the same author, which may be widely separated in time ("`books`" task). The dataset comprises $1,019$ authors, $2,286$ books and $527,340$ pages and their binarised counterparts, spanning at least 500 years.

Keywords: Writer Identification · Historical Document Retrieval · Handwriting Analysis

1 Introduction

Traditional frameworks for writer identification based on handcrafted features, ranging from early [7,5,2,19,3,9] to the most recent and advanced approaches [22,11,23,21,6], typically treat handwriting style as a static snapshot of an individual at a specific point in time. In such settings, the identity of a writer is represented through a limited set of pages captured within a narrow temporal window. While effective for short-term pattern recognition, this paradigm overlooks a crucial dimension of handwriting analysis: the evolution of writing style over time.

This simplification limits both the interpretability and the real-world applicability of handwriting-based author identification, particularly in historical

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Fig. 1. Visual abstract of the presented challenge. Practitioners are expected to, given a set of queries (entire full-resolution books), retrieve all instances of manuscripts in the dataset that could be potentially written by the same author through their life.

contexts. From a palaeographic perspective, handwriting naturally evolves due to ageing, context, and changing writing materials. Consequently, models trained on temporally constrained samples fail to generalize across an author's lifetime or to establish long-term stylistic continuities.

The AnyScript competition introduces a new benchmark designed to evaluate systems on this temporal transitivity of authorship. In a query-by-example setting, participants must retrieve and associate manuscript pages authored by the same individual across different years or even decades (See Fig.1). This setup encourages the development of models capable of distinguishing persistent stylistic traits from temporal variations. Moreover, the competition goes beyond single-page retrieval by requiring a holistic view of author identification, where relevant handwriting features must be selectively integrated across multi-page manuscripts from multiple domains: handwritten text in letters, plays and other texts as well as music scores. Because it is quite a demanding task to directly in-

fer a common authorship from manuscripts very separated in time (say, relation A $\overset{\texttt{same author}}{\longleftrightarrow}$ C), we expect methods to incorporate the notion of *temporal reasoning* through intermediate, and easier, association (i.e. through a manuscript *B* which is temporally closer to both manuscripts *A*and *C*: A $\overset{\texttt{same author}}{\longleftrightarrow}$ B, B $\overset{\texttt{same author}}{\longleftrightarrow}$ C, therefore, A $\leftrightarrow$ C).

To summarize, the main contributions of this competition and the proposed AnyScript dataset are the introduction of a novel benchmark for temporally-aware, multi-page, and multi-domain writer identification and the promotion of methods that incorporate reasoning over all of these dimensions. This report summarises the work of the participants and their results under this setting and is structured as follows. In Sec. 2, we introduce the AnyScript dataset. Sec. 3 presents the competition framework, including the evaluation metrics and the competition timeline. This is followed by Sec. 4 where a description of the various methods proposed by the authors to address the challenge is presented. Finally, the results and conclusions of the competition are presented in Sec. 5 and Sec. 6.

2 The AnyScript Dataset

The competition data set, as exemplified in Figure 2, is built from a scraping of the Spanish Digital Library [1]. All records labeled as handwritten documents that had available digitized copies were collected, regardless of their type. For an important number of these documents, the author of the archived copy is not known, and in those cases where it is, the information is not stored systematically in the database (it can *sometimes* be found in prose within the comments of the record, but it is often uncertain, unreliable or difficult to extract). Therefore, for writer identification tasks we have chosen only to include autograph copies of works that is, copies produced by the original creator of said works, as it is easier to guarantee the identity of the author. Even then, some elements labelled as autographs in the library date after the death of their authors. We have chosen to remove these documents, given than in most cases they appear to be partial or total forgeries from older manuscripts. Each document has been binarised used Sauvola and Niblack algorithms based on the domain of the document.

The final version of the dataset contains around 527,340 distinct images and their binarized counterparts from 2,286 different volumes written by 1,019 unique hands. It contains copies of novels, plays, music scores, and other literary and artistic works between the 18th and the 20th centuries, most of them in Spanish (See Fig.2).

3 Competition Framework

The competition is divided in two tasks, each of which tackles two different, yet complementary, writer identification problems, which are described below. We also provide a description of the metrics and the schedule of the competition.

[1] *Biblioteca Digital Hispánica*, https://bdh.bne.es/

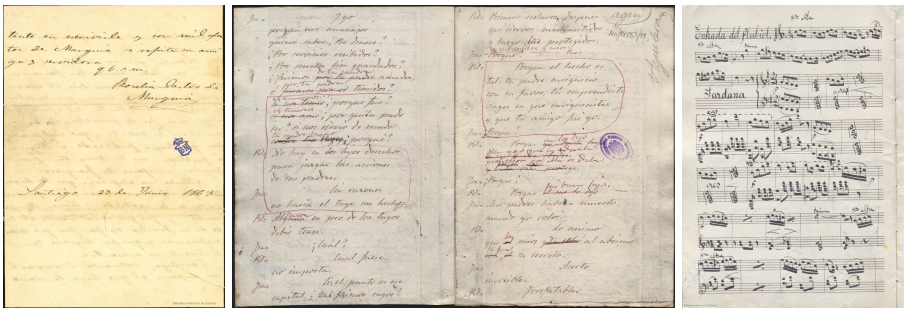

Fig. 2. Examples of personal letters, theater scripts and music scores found in the dataset.

3.1 Task A: `pages`

In this setting, participants are provided with individual pages that were excluded from books in the training set. The goal is to identify the books where these pages belong. Retrieval is evaluated at the **page level**, and each retrieved ID represents a single page. In more specific terms, participants receive a query-by-example test set consisting on $1,000$ of pages and they have to return, for each page, the similarity score against all other samples in the training set.

3.2 Task B: `books`

In this track, participants are provided with complete books that were excluded from the training set. The objective is to identify other books authored by the same individual, possibly separated by long time spans. Retrieval is evaluated at the **book level**, where each retrieved ID represents a whole volume. In short, participants receive a query-by-example test set consisting on 200 of books and they have to return, for each input book, the similarity score with each sample in the training set.

3.3 Construction of the Test Set

The test set for the AnyScript competition consists on $8,247$ queries excluded from the training data. This means that the objective of the competition is to retrieve documents which correspond to either volumes or authors already present in the training set.

For the `books` track we sample 87 random text books, and 8 books of music scores, totalling $7,667$ pages. These books are sampled following a distribution inversely proportional to the amount of documents available in bins of 50 years. This way, as shown in Figure 3, the temporal distribution in the test set is balanced. These books are removed from the training set, and it is ensured that the selected books belong to authors that have at least two books in the full

Table 1. Books / Pages for the queries within each test track and document type

	`pages`	`books`
Handwritten Text	78 / 491	87 / 6,672
Music Scores	12 / 89	13 / 995

dataset. This guarantees that there will be at least one other book from the same author and thus that retrieval within the training set is possible.

For the `pages` track we use the same methodology to sample books from the remainder of the training set. However, instead of fully removing the selected books and using them as test queries, we use half of their pages as queries and leave the rest in the training set to perform retrieval. The breakdown of these splits can be found in Table 1

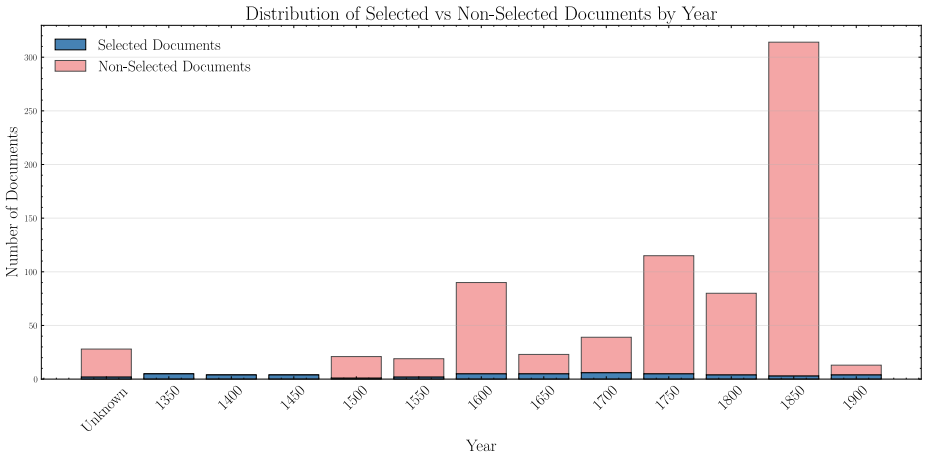

Fig. 3. Books on the test split are sampled with a balanced distribution with respect to the temporal axis.

3.4 Evaluation

As is customary in writer identification research, the proposed tasks are evaluated under the framework of a query-by-example retrieval problem. Given a query document (a page or a book), systems are required to retrieve other documents written by the same author from the dataset. The evaluation relies on two complementary metrics: Recall@K and Mean Average Precision (mAP).

Recall@K. Recall at K (for $K = 1, 5, 10 \dots$) measures the proportion of relevant documents (i.e., those written by the same author as the query) that appear

among the top K retrieved items:

$$\text{Recall@}K = \frac{\text{Number of relevant documents in top } K}{\text{Total number of relevant documents}}. \tag{1}$$

Mean Average Precision (mAP). Mean Average Precision summarizes the ranking performance by computing the average precision for each query and then averaging across all queries. It rewards systems that rank relevant items higher in the retrieval list:

$$\text{AP}(q) = \frac{1}{R_q} \sum_{k=1}^{N} P_q(k) \cdot \text{rel}_q(k), \quad \text{mAP} = \frac{1}{Q} \sum_{q=1}^{Q} \text{AP}(q), \tag{2}$$

where R_q is the total number of relevant documents for query q, $P_q(k)$ is the precision at rank k, and $\text{rel}_q(k)$ indicates whether the document at rank k is relevant (1) or not (0).

3.5 Schedule

The competition started at the end of January with the release of the test set. On March 15th, the validation platform was opened and remained available until the submission deadline on April 17th. During this period, the progress of the different participants can be observed in Fig. 4.

4 Methods

Participants have submitted methods that address the challenge of the competition in fairly diverse ways, although all methods are variations of the standard, Deep Learning-based image retrieval pipeline: some form of visual feature extraction backbone tuned with metric learning or a classification head. The only partial exception to this rule is the `books` track submission for Team A, which instead uses a graph-based approach built from SIFT and ResNet features across all pages of the dataset.

The feature extractors that have been used throughout the competition are the Swin Transformer [17], VGG [25], DINOv3 [24] and Xception [4] architectures, not showing any preference over CNN or Transformer-based models. Each team has chosen to refine these backbone features using different techniques. We have found NetVLAD [1] to be the only component that has been reused by multiple competitors (Team A and Team D). Team C instead chose to use layers from an AMD-HookNet++ [28] — which extends a U-Net architecture with attention layers — whereas Team B used a stack of standard Transformer layers for feature aggregation.

Some teams have built adjustments that target the specificities of each of the document families within the dataset. Team A's `pages` submission feeds the model with some page-level properties such aspect ratio to provide the model

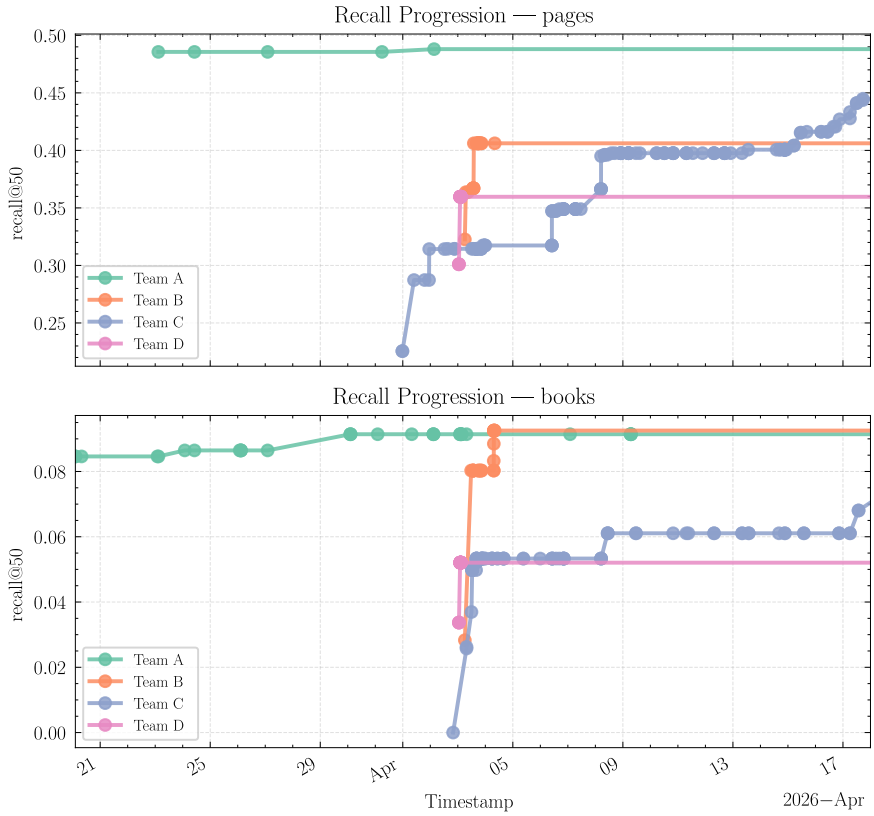

Fig. 4. Progress of submissions during the competition window for the `pages` and `books` tracks respectively previous to the validation of the results..

with some broad context. Moreover, their `books` track method incorporates a Hough transform step to remove staff lines for music scores, which can affect their SIFT-based methodology. Team B's method uses OCR to remove potential handwriting line false positives, which they claim affects their capacity of handling Music Scores and empty pages despite the fact they got good results there nonetheless.

In the following subsections we show each participant team's full method.

4.1 Team A

Participants: Zhonghao Shen, Wentao Cui, Xunhui Qin
Affiliation: Chongqing I-Signing Digital Technology Co.,Ltd., China
Description of the Method:
Pages track method: We introduce a metadata-aware framework for historical book retrieval that explicitly injects document metadata as a structural prior

into feature learning pipeline, learning discriminative page-level representations sensitive to both visual content and document structure. Given an input historical book page, we first partition the page into a sequence of image patches while explicitly preserving the original document metadata. Such metadata includes the page size, aspect ratio, layout type, and the spatial attributes of each patch, such as its position, scale, and relative location within the page as an explicit and informative signal that guides representation learning.

Each patch is encoded by a Swin Transformer [17] backbone to capture fine-grained visual patterns from the page content. In parallel, the corresponding metadata is embedded into a structural representation and fused with the visual tokens, enabling the backbone to become metadata-aware at the patch level. The patch features are then aggregated by an attention-based NetVLAD module to assign higher weights to patches that are more informative for retrieval. Furthermore, the aggregated representation is fused with page-level metadata to produce a compact and robust page-level descriptor that jointly encodes content semantics and layout-aware structural information.

On top of the fused page representation, we adopt a multi-task learning strategy with two complementary branches. The classification branch involves three independent learning objectives (page, book, and writer) and employs Focal Loss [16] to mitigate class imbalance and enhance the discriminability of hard samples. The representation branch is designed for retrieval and metric learning to shape the embedding space [12]. We also design data augmentation strategies during training. Specifically, for two-page layouts, we randomly crop either the left or right page as input to increase sample diversity. Additionally, we construct positive pairs within the book-including random, adjacent-page, and first/last-page pairings-to improve retrieval accuracy for various page layouts within the same book.

Books track method: During preprocessing, every page of its binary version is resized to 1024×2048, with height being rescaled and width being padded or cropped accordingly. We observe that some pages are multi-paged, which suggests huge width disparity, so an image mask is kept to record patched positions. No data augmentation is used at all. Then, all contours [20] are tracked and filtered based on their sizes, compactness and irregularity; contour points are resampled to be evenly spaced and counter-clockwise. For those music score manuscripts that might mess up with handwritings, Hough line transform [10] is utilized to erase music lines beforehand.

We characterize each page as a graph rather than an image. The graph's nodes are described by sequence features, SIFT [18] and deep image features. Sequence features include x, y, angle and their derivatives along its contour path. SIFT is conducted based on gray-scale image yet with no angle self-adaptation; deep image features are the feature maps learned by the Resnet-34's first layer. Those features are concatenated along feature dimension. Graph's edges are build up in two ways. The first type simply connects each node to its two neighbours along its contour path; the second type finds k nearest points of other contours (one point per contour).

The model's backbone is composed of stacks of GATs [26]. The learned feature map is fed to NetVLAD [1] with 64 clusters to obtain the ultimate feature vector along with a Squeeze-Excitation Pooling layer after the feature map to learn the importance of each nodes and it replaces the padding mask in baseline. Finally, a FCN layer is used to predict to which writer each page belongs.

4.2 Team B

Participants: Jan Kohút, Marek Vaško, Martin Kišš, Michal Hradiš
Affiliation: Faculty of Information Technology, Brno University of Technology, Brno, Czech Republic
Description of the Method: Our pipeline includes a preprocessing step for line extraction and a line embedding network for feature extraction. Lines are extracted using a segmentation-based layout model [14], and then we apply OCR for additional line filtering [15]. Filtering is transcription confidence-based. For lines up to 5 characters long, a transcription confidence threshold of 0.8 is used. Lines up to 8 characters are filtered with a threshold of 0.7, and any longer lines are filtered with a threshold of 0.6. We also filtered out lines containing a watermark.

The line feature extractor uses a VGG-based backbone with transformer layers for feature aggregation. We use initial weights from a network pre-trained for OCR [15]. We train two versions of the feature extractor network using Proxy Anchor Loss [13]; one for the book-level task, where ground truths are author ids, and another for the page-level task, where ground truths are book ids. We reserved 2 pages per book for the testing dataset and used book retrieval mAP for the ablation study. The best training configuration used a learning rate of 2×10^{-4} (halved after 100k iterations), a batch size of 200, 268k iterations on the page-level task, and 176k iterations on the book-level task. We used early stopping for both tasks, where the monitored metric was book/author retrieval mAP.

Training data is augmented using a standard set of augmentations, including color and brightness adjustments, contrast enhancement methods such as CLAHE and histogram equalization, various blur and sharpening filters, edge detection, and photometric distortions, including solarization, posterization, and grayscale conversion. The main limitation of our approach is its inability to handle empty pages and music sheets, due to its line-based OCR. *Page-level adaptations:* Each page is represented by a set of text-line embeddings. For a given query page, we compute its similarity to every training page as the mean cosine similarity across all pairs of their text lines. The resulting per-training-page scores are smoothed toward the training book's mean via $s' = 0.25 \cdot s_{\text{page}} + 0.75 \cdot \bar{s}_{\text{book}}$.

Book-level adaptations: Per-training-page similarities are computed for each query page in the same way (without alpha smoothing). Once all pages of a query book have been processed, the per-training-page scores are averaged across the query book's pages (mean pooling), yielding a single similarity score per training page.

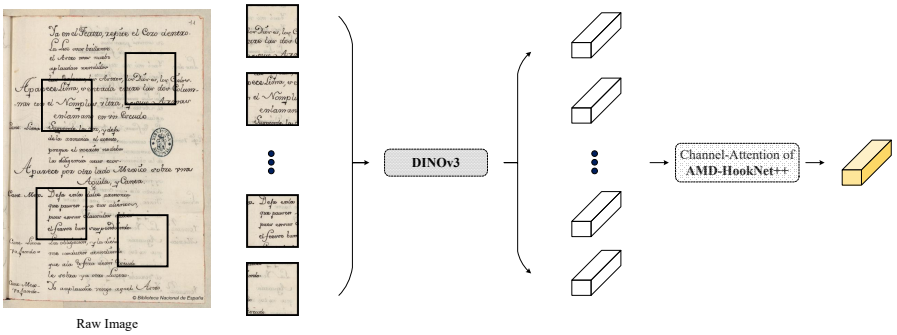

Fig. 5. Overall framework of the approach of Team C.

4.3 Team C

Participants: Fei Wu, Vincent Christlein
Affiliation: Pattern Recognition Lab, Friedrich-Alexander-Universität Erlangen-Nürnberg
Description of the Method: We adopt the same architecture for both competition tracks, i.e., for book retrieval and page retrieval. The only difference lies in the number of classes used in the loss function during training. For each training iteration, we randomly sample 16 patches (RGB-based) per page as input to the DINOv3 [24] backbone, which is pretrained on the LVD-1689M dataset. Each patch is 256 × 256 pixels (see Figure 5). The resulting 16 feature maps are then aggregated into a single unified feature embedding using the channel attention mechanism from AMD-HookNet++ [28]. After ℓ_2-normalization, this embedding represents the holistic information of the entire page.

Our model is trained using the Multi-Similarity Loss LMS [27], which is defined as follows:

$$\mathcal{L}_{MS} = \frac{1}{m}\sum_{i=1}^{m}\left\{\frac{1}{\alpha}\log\left[1+\sum_{k\in\mathcal{P}_i} e^{-\alpha(s_{ik}-\lambda)}\right] + \frac{1}{\beta}\log\left[1+\sum_{k\in\mathcal{N}_i} e^{\beta(s_{ik}-\lambda)}\right]\right\} \quad (3)$$

For the `books` track, the number of classes w.r.t. the loss function is set to the number of unique authors (1019), whereas for the `pages` track, it is set to the number of books (2286). We use stochastic gradient descent (SGD) as the optimizer, with momentum and weight decay set to 0.9 and 1e-4, respectively. The learning rate decays exponentially from 1e-2 to 1e-4. The model is trained for 150 epochs with a batch size of 32. For each class, we randomly sample two pages to construct the positive pairs and negative pairs for the metric learning. Within each page, 16 patches are randomly extracted. The input image patch size for DINOv3 is 256×256 pixels, and no data augmentation is applied during training. The hyperparameters α, β, and λ in Eq. 3 are set to 2.0, 70, and 0.5,

respectively. We use the LVD-1689M pretrained DINOv3 as the backbone, while the channel attention mechanism from AMD-HookNet++ [28] is employed for feature aggregation. Note that all embeddings in the loss function (Eq. 3) are ℓ_2-normalized. The experiments are performed on a server equipped with an AMD EPYC 7713@2.0 GHz CPU and a single Nvidia A100-SXM4 GPU.

4.4 Team D

Participants: Peirong Zhang, Lianwen Jin
Affiliation: South China University of Technology, Guangzhou, China
Description of the Method: For the historical handwritten book page retrieval task, we propose a page representation learning framework that combines Wavelet Convolution (WTConv) [8], the Xception backbone [4], and NetVLAD [1]. Given a book page image as input, the WTConv module is introduced to enlarge the receptive field with modest computational overhead, enabling the model to capture broad page-level layout cues across lines and paragraphs. In parallel, Xception serves as the backbone network to extract fine-grained local handwriting and texture patterns. To learn discriminative representations, the model is trained with two auxiliary classification objectives, namely author identity classification and book affiliation classification, and the network parameters are jointly optimized in a multi-task learning manner. The feature representation produced by the model is then fed into a NetVLAD module for aggregation, yielding a compact and robust page-level descriptor. Finally, the aggregated representation is used for similarity-based retrieval during testing.

5 Results

We present the overall results of the competition in Table 2, logging both mAP@50 and Recall@50 metrics for both tracks across all submissions. It should be mentioned that, since Recall is computed using only the top 50 retrieved elements for each query, it is not possible to obtain all of the relevant elements for any

Table 2. Final results for the competition. Results for each team are shown both separately and jointly for text and score-based documents and also split across tracks. We log mAP@50 and Recall@50, while also providing the theoretical maximum recall on each scenario for context. Best and second-best results in bold and underlined respectively.

		Team A			**Team B**			**Team C**			**Team D**		
		Scores	Text	Total	Scores	Text	Total	Scores	Text	Total	Scores	Text	Total
Pages	mAP@50	0.967	0.960	**0.961**	0.749	0.778	0.775	0.904	0.910	<u>0.909</u>	0.895	0.846	0.854
	Recall@50	0.329	0.572	**0.488**	0.241	0.431	0.406	0.293	0.530	<u>0.464</u>	0.256	0.410	0.359
	Max Recall@50	*0.381*	*0.669*	*0.625*	*0.381*	*0.669*	*0.625*	*0.381*	*0.669*	*0.625*	*0.381*	*0.669*	*0.625*
Books	mAP@50	1.000	0.824	**0.841**	1.000	0.811	<u>0.829</u>	1.000	0.641	0.676	0.833	0.596	0.619
	Recall@50	0.025	0.098	<u>0.091</u>	0.025	0.099	**0.092**	0.025	0.056	0.053	0.020	0.055	0.052
	Max Recall@50	*0.025*	*0.160*	*0.147*	*0.025*	*0.160*	*0.147*	*0.025*	*0.160*	*0.147*	*0.025*	*0.160*	*0.147*

given query in one go, as in some cases there are hundreds of potential positives. Thus, to fully contextualise the results, we have also provided the maximum obtainable recall in each of the cases (scores, text and overall) for both tracks. In turn, this setting technically makes the competition easier, as it is possible to log the straightforward samples first and leave the ambiguous pages outside of the range considered by the metrics.

The results reveal distinct behaviours between the page-level and book-level tracks, highlighting some of the challenges posed by temporally-aware writer identification. On the `pages` track, all methods achieve high mAP@50 values, with Team A obtaining a mAP@50 of 0.961, followed closely by Team C at 0.909. Given that the task involves finding elements belonging to a set of relatively homogeneously-looking images, it is not surprising that most methods manage to obtain very good scores on this track. Studying the queries that tend to fail, as seen in Fig. 6 and Fig. 7 we can conclude that the failure cases can be mostly attributed to query images that are very different from the rest of the containing book, further confirmed by the fact that most models react similarly.

In the `books` track there seems to be a significant drop in performance in Recall@50 on all methods, with most of them falling significantly below the theoretical maximum, while still reaching high mAP@50 values. This apparent discrepancy can be explained by the large number of relevant — yet visually distinct — documents associated with each query in this setting. Team A got the best overall results in mAP but Team B method got slightly better results on Recall.

Delving a bit deeper on the overarching theme of writer identification across time one can observe in Figure 8 a general prevalence of methods to showcase superior performance in later, more modern documents. This is the exception of Team A, for which more modern document achieve worse performance with a particular advantage in early 16th century documents (see Figure 9, `pages`). We can partially attribute it to the fact that Team A has used a loss designed to deal

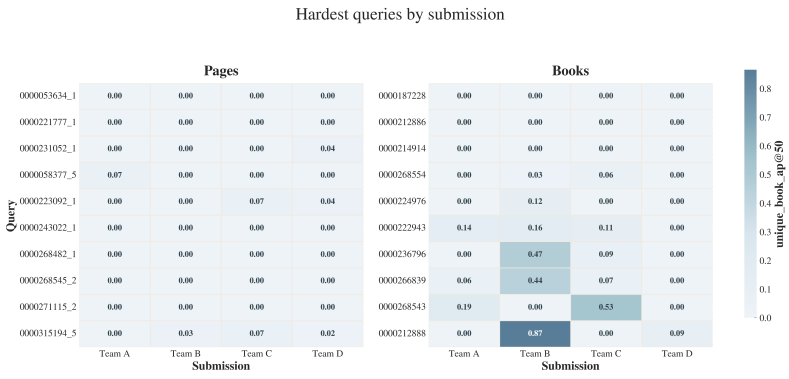

Fig. 6. Agreement on the hardest queries both in `books` and `pages` on mAP@50.

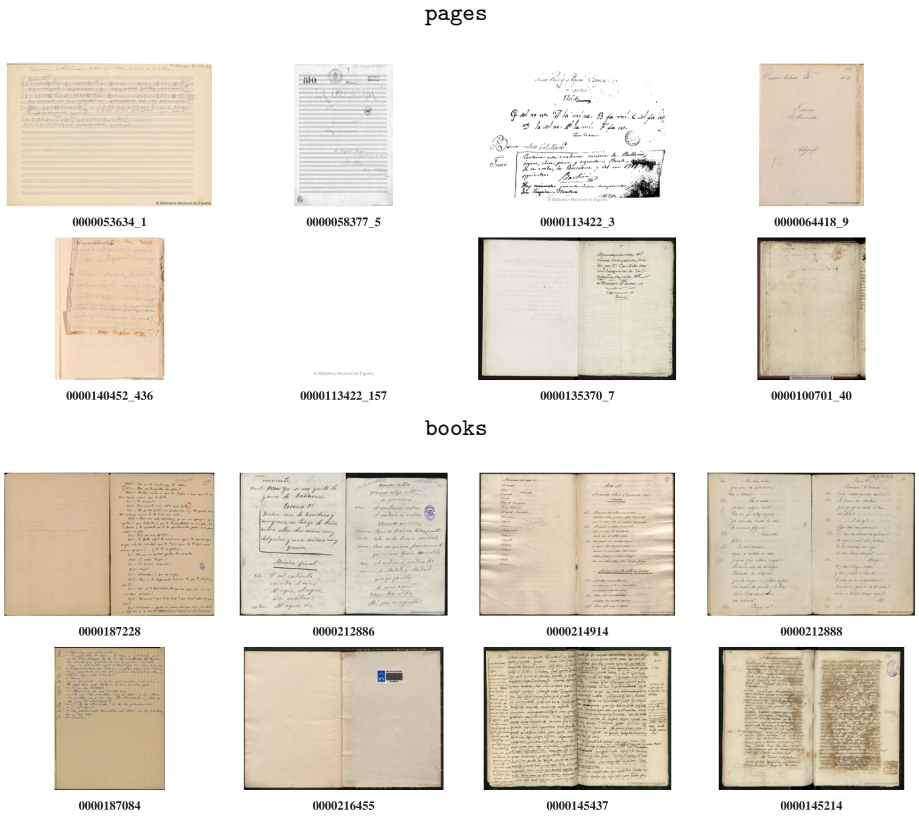

Fig. 7. Examples of the hardest queries within the dataset in both the `pages` and the `books` tracks.

with class imbalance and, in part, this should favour more diversely represented embedding spaces.

Additionally, a consistent performance gap is observed between the Scores and Text subsets, with higher results typically obtained on Scores. This is unsurprising, as scores are a very unique type of document that contains easily distinguishable graphical features. As a matter of fact, we have observed almost all submissions in both tracks are homogeneous in retrieved document typology, with the exception being a few queries on the `pages` track that were actually scans of score book bindings. For the `books` track, on the other hand, we have observed perfect results for most teams, highlighting that there is probably not enough of a challenge in the proposed setting. In particular, most score books are very lengthy and their internal variability is rather small, which makes it relatively easy to pick pages that belong to the same author as long as there exists one score book that is close enough to the query.

Overall, the results support the central premise of the benchmark: while methods perform well under short-term or local similarity assumptions, they still

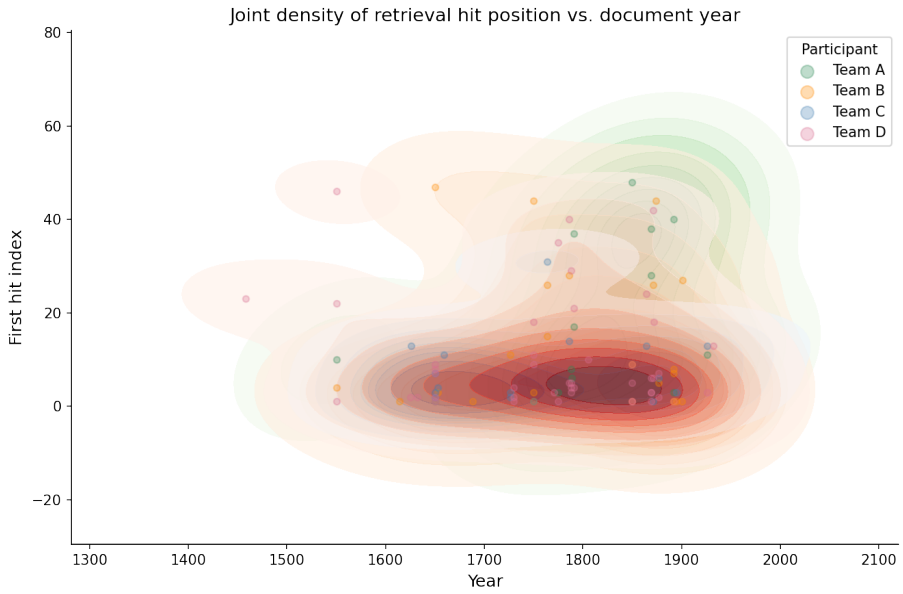

Fig. 8. Date-sensitivity of every method. Generally speaking, all methods perform the best in 19th-20th century documents. However, Method A particularly fails in correctly predicting samples for such period; in their case, the method performs bests in 16th-18th century documents.

face challenges in capturing long-term stylistic continuity, as most methods did well on the `pages` track but fared worse on the `books` track. This underscores the need for approaches that explicitly model handwriting evolution and incorporate temporal reasoning mechanisms, as intended by the design of the AnyScript dataset.

6 Conclusion

The competition got a final set of four contenders on each track, with Team A leading comfortably on the `pages` track and Team B scoring a tremendously narrow win over Team A in Recall@50 on the `books` track. Overall, the results indicate that current models perform well under retrieval scenarios where the queried document is within the same period or the same domain as the targets, but still struggle when it comes to finding works from the same author separated by a large temporal or stylistic gap. The take-home message is that the paradigm introduced in this competition for long-term handwriting author identification is challenging and there is still margin for improvement.

Most submissions focused on the extraction of stylistic features from the images and then performing either classification or metric learning to learn image embeddings from which to perform retrieval. This is unsurprising, as it is

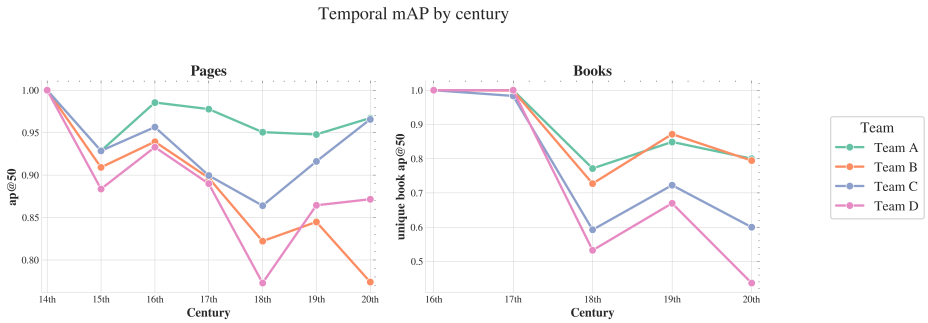

Fig. 9. Temporal sensitivity of each method w.r.t. the century of the query image both in the `pages` and the `books` tracks.

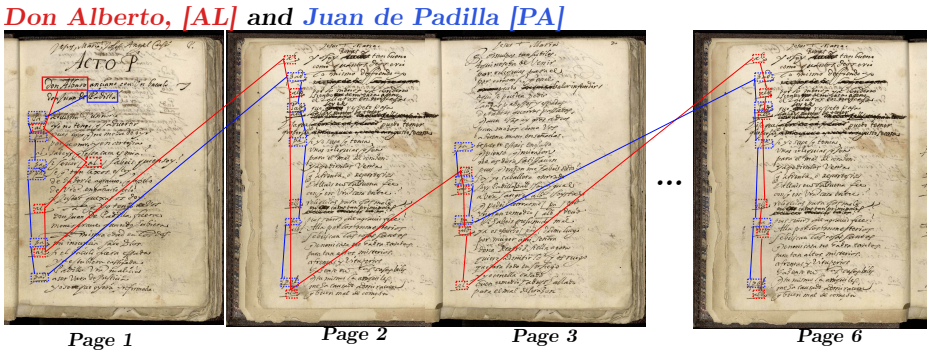

Fig. 10. Example of a book from the AnyScript dataset for which page associations can be established without relying on stylistic features. Semantic continuity, illustrated here by the recurring interaction between Don Alberto and Juan de Padilla, allows linking non-consecutive pages. The colored lines indicate correspondences identified through content rather than visual style.

the dominant paradigm for these kinds of tasks and allows re-usage of practical knowledge and previously trained components. However, as shown in Fig. 10, there are interesting shortcuts — or consistency clues, rather — that we hypothesise could have been used to identify pages of the same work without relying purely on visual features. In particular, for document typologies that contain recurrent fragments of text such as theatre, we believe that it should be possible to find all pages belonging to the same work by finding these repeating patterns. Even for novels or scores, there are content-based clues — *leitmotifs* or writing styles — that seem like strong priors in retrieval and which we would like to see being explored in the future.

We would also like to expand the collection of samples to include newer domains and also increase the difficulty of the existing ones. In particular, the

proposed setting for music scores seems to be solved fairly well by most methods, thus being a prime target for improvement on the future.

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